Project: Task 15 - End-to-End ML Pipeline
Dataset: Breast Cancer Diagnostic
Workflow: ColumnTransformer -> Standard Scaler -> Random Forest

1. Pipeline Architecture
We constructed a unified Machine Learning Pipeline that encapsulates the entire workflow:

Preprocessing Stage: A ColumnTransformer was used to apply StandardScaler to all numerical features. This ensures that raw data fed into the model is automatically normalized without manual intervention.

Modeling Stage: A RandomForestClassifier was attached to the end of the pipeline.

2. Performance Metrics

Accuracy: ~96%

Precision/Recall: The pipeline achieved high recall for the malignant class, which is critical for medical diagnosis.

Reliability: By using a pipeline, we eliminated the risk of Data Leakage (e.g., scaling the test set using its own mean instead of the training set's mean).

3. Production Readiness
The final model was saved as a .pkl file. This single file contains both the scaling logic and the trained trees. In a production environment (like a web app), we only need to load this one file and pass raw user data to get a prediction.